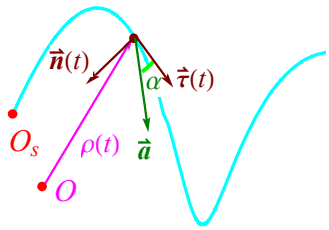
Homework 01, Week 02 2023.09.11 Due 2023.09.18 L01.06, L01.07, L01.08, L01.16

★ L01.06

质点在平面上运动，速度矢量与加速度矢量的夹角 α 保持不变．试证，在此情况下，$v\,e^{-\theta\cot\alpha}=$ 常数，这里v是速率，θ 是轨道切线的方位角．

Solution:

如图



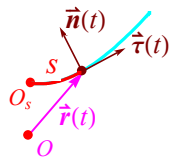
$$\begin{cases} a_\tau = \dot{v} \\ a_n = \dfrac{v^2}{R} \end{cases} \Longrightarrow \frac{a_\tau}{a_n} = \cot\alpha \Longrightarrow \frac{R\,\dot{v}}{v^2} = \cot\alpha$$

$$\Longrightarrow \cot\alpha = \frac{R}{v^2}\frac{dv}{ds}\frac{ds}{dt} = \frac{R}{v^2}\frac{dv}{d\theta}\frac{d\theta}{ds}v = \frac{R}{v}\frac{dv}{d\theta}\frac{1}{R} \Longrightarrow \frac{dv}{d\theta} = v\cot\alpha \Longrightarrow v = C\,e^{\theta\cot\alpha} \Longrightarrow v\,e^{-\theta\cot\alpha} = \text{常数}$$

★ L01.07

质点在平面上运动，速度的 x 分量 v_x 保持不变，试证加速度 $a=\dfrac{v^3}{v_x R}$，其中 R 为曲率半径

Solutions



$$\begin{cases} a_\tau = \dot{v} \\ a_n = \dfrac{v^2}{R} \end{cases} \Longrightarrow a^2 = \left(\frac{v^2}{R}\right)^2 + \dot{v}^2$$

$$v_x = v\cos\theta = \text{const.} \xrightarrow{\text{两边对 } t \text{ 求导}} 0 = \dot{v}\cos\theta - v\dot{\theta}\sin\theta \Longrightarrow \dot{v} = v\dot{\theta}\tan\theta$$

$$\dot{\theta} = \frac{d\theta}{dt} = \frac{d\theta}{ds}\frac{ds}{dt} = \frac{v}{R} \Longrightarrow \dot{v} = \frac{v^2}{R}\tan\theta$$

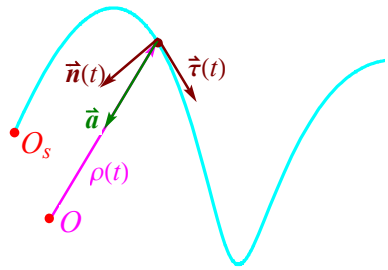
$$a^2 = \left(\frac{v^2}{R}\right)^2 + \left(\frac{v^2}{R}\right)^2\tan^2\theta = \left(\frac{v^2}{R}\right)^2\sec^2\theta \Longrightarrow a = \frac{v^2}{R\cos\theta} = \frac{v^3}{R\,v\cos\theta} = \frac{v^3}{R\,v_x}$$

★ L01.08

质点作平面运动，其加速度矢量 $\vec{a}$ 始终通过某个定点 O．试证 $a=v\dfrac{dv}{d\rho}$，其中 ρ 是质点与 O 的距离．

Solution I:

如图



$$a_\rho = \ddot{\rho} - \rho\dot{\phi}^2$$

$$a_\phi = \rho\ddot{\phi} + 2\dot{\rho}\dot{\phi}$$

依题意：$a_\phi = \rho\ddot{\phi} + 2\dot{\rho}\dot{\phi} = 0$ (1)

速度：$v = \dot{\rho}\vec{e}_\rho + \rho\dot{\phi}\vec{e}_\phi \Longrightarrow v^2 = \dot{\rho}^2 + \rho^2\dot{\phi}^2$

$$\frac{d\,v^2}{d\rho} = \frac{d\left(\dot{\rho}^2 + \rho^2\dot{\phi}^2\right)}{d\rho} = 2\dot{\rho}\frac{d\dot{\rho}}{d\rho} + 2\rho\dot{\phi}^2 + 2\rho^2\dot{\phi}\frac{d\dot{\phi}}{d\rho}$$

利用：$\dfrac{d\dot{\rho}}{d\rho}=\dfrac{d\dot{\rho}}{dt}\dfrac{1}{\dfrac{d\rho}{dt}}=\dfrac{\ddot{\rho}}{\dot{\rho}}$，$\dfrac{d\dot{\phi}}{d\rho}=\dfrac{d\dot{\phi}/dt}{d\rho/dt}=\dfrac{\ddot{\phi}}{\dot{\rho}}$ $\Longrightarrow$ $\dfrac{d\,v^2}{d\rho}=2\left[\ddot{\rho}+\rho\dot{\phi}^2+\rho^2\dot{\phi}\dfrac{\ddot{\phi}}{\dot{\rho}}\right]$，

故：$\dfrac{1}{2}\dfrac{d\,v^2}{d\rho}=\left[\ddot{\rho}+\rho\dot{\phi}^2+\rho^2\dot{\phi}\dfrac{\ddot{\phi}}{\dot{\rho}}\right]$ 利用 (1) 式

$$=\left[\ddot{\rho}+\rho\dot{\phi}^2+\rho\dot{\phi}\frac{1}{\dot{\rho}}\left(-2\dot{\rho}\dot{\phi}\right)\right]=\ddot{\rho}-\rho\dot{\phi}^2=a_\rho=a$$

即：$\dfrac{1}{2}\dfrac{d\,v^2}{d\rho}=a$ $\Longrightarrow$ $a=v\dfrac{dv}{d\rho}$

Solution II：

看 $v\,dv=\dfrac{1}{2}dv^2=\dfrac{1}{2}d(\vec{v}\cdot\vec{v})=\vec{v}\cdot d\vec{v}=(\vec{v}\,dt)\cdot\dfrac{d\vec{v}}{dt}=(d\vec{r})\cdot\vec{a}=\vec{a}\cdot d\vec{r}$

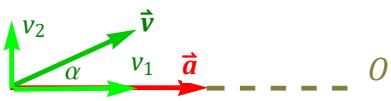
对二维且加速度矢量 $\vec{a}$ 始终通过某个定点 O，则：$\vec{a}=a_\rho\vec{e}_\rho$

$\vec{a}\cdot d\vec{r}=\vec{a}\cdot d\vec{\rho}=a_\rho\vec{e}_\rho\cdot(\vec{v}\,dt)=a_\rho\vec{e}_\rho\cdot\left(\dot{\rho}\vec{e}_\rho+\rho\dot{\theta}\vec{e}_\theta\right)dt=a_\rho\dot{\rho}\,dt=a_\rho\,d\rho$

但 $\vec{a}\cdot d\vec{r}=v\,dv$

联立：$a_\rho\,d\rho=v\,dv$ $\Longrightarrow$ $a_\rho=v\dfrac{dv}{d\rho}$

Solution III



$$\begin{cases} a=a_1=\dfrac{d\,v_1}{dt}=\dfrac{d(v\cos\alpha)}{dt} \\ a_2=0=\dfrac{d\,v_2}{dt}=\dfrac{d(v\sin\alpha)}{dt} \end{cases}$$

故：$\begin{cases} a=\dot{v}\cos\alpha-v\dot{\alpha}\sin\alpha \\ 0=\dot{v}\sin\alpha+v\dot{\alpha}\cos\alpha \end{cases}$ $\Longrightarrow$ $a\cos\alpha=\dot{v}=\dfrac{dv}{dt}$ (1)

由自然坐标加速度公式 $\vec{a}=\dot{v}\vec{\tau}+v\dfrac{d\vec{\tau}}{dt}=\dot{v}\vec{\tau}+\dfrac{v^2}{R}\vec{n}$ 也可得到 $a\cos\alpha=\dot{v}$

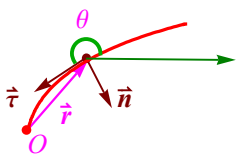
又，由极坐标速度公式：$\vec{v}=\dot{\rho}\hat{e}_\rho+\rho\dot{\phi}\hat{e}_\phi$ $\Longrightarrow$ $v\cos\alpha=\dot{\rho}=\dfrac{d\rho}{dt}$ (2)

(1)/(2) 即得：$\dfrac{a}{v}=\dfrac{dv}{d\rho}$ $\Longrightarrow$ $a=a_1=a_\rho=v\dfrac{dv}{d\rho}$

★ L01.16

质点沿光滑抛物线 $y^2=2x$ 无初速滑下，质点初始坐标为 (2, 2)。问质点在何处脱离抛物线

Solution



在脱离抛物线的点 $\left(x_0,\sqrt{2x_0}\right)$：

$a_n=\dfrac{v^2}{R}=g\cos\theta$，$\tan\theta=y'=\dfrac{1}{\sqrt{2x_0}}$

$\dfrac{1}{2}m\,v^2=m\,g\left(2-\sqrt{2x_0}\right)$ $\Longrightarrow$ $v^2=2g\left(2-\sqrt{2x_0}\right)$

曲率半径： $R = \dfrac{ds}{d\theta} = \dfrac{\sqrt{(dx)^2 + (dy)^2}}{d\tan\theta}(\tan\theta)' = \dfrac{\sqrt{1 + \dfrac{1}{2x_0}}}{\dfrac{1}{2\sqrt{2x_0^3}}}\sec^2\theta = (2x_0 + 1)^{3/2}$

故： $\dfrac{v^2}{R} = g\cos\theta \implies \dfrac{2\left(2 - \sqrt{2x_0}\right)}{(2x_0 + 1)^{3/2}} = \dfrac{1}{\left(1 + \dfrac{1}{2x_0}\right)^{1/2}}$

$\xRightarrow{\sqrt{2x_0} = y_0} \quad y_0^3 + 3y_0 - 4 = 0 \implies \begin{cases} y_0 = 1 \\ x_0 = 1/2 \end{cases}$